

Lecture 6: Atmospheres II

Clouds, Weather, & Climate

Planetary Systems

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Learning objectives

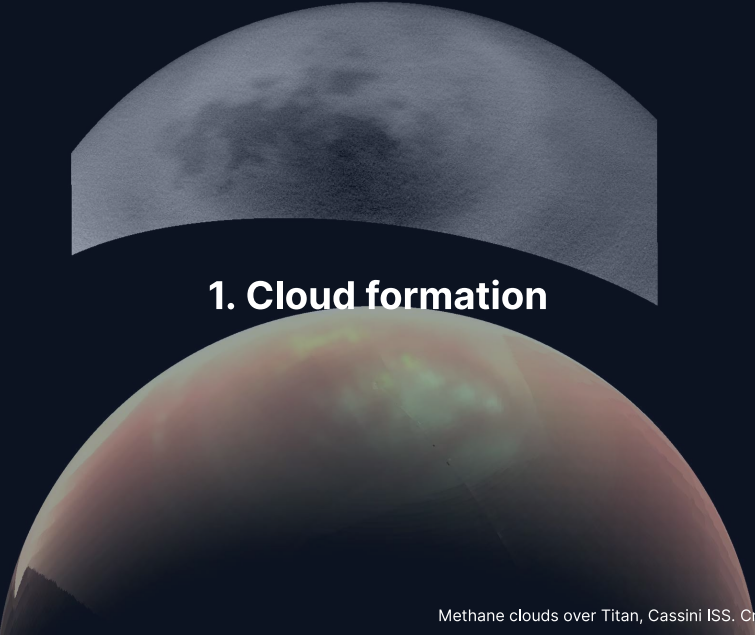
By the end of this lecture, you will be able to:

- Derive the Clausius-Clapeyron equation and predict where clouds form
- Explain circulation with the Coriolis effect, the Rossby number and geostrophic balance
- Explain the faint young Sun paradox and the carbonate-silicate thermostat

Recap: Lecture 5

- Hydrostatic equilibrium $dP = -\rho g dz$ and scale height $H = k_B T / (\mu m_u g)$
- Greenhouse warming: $T_s > T_{\text{eq}}$
- Convection sets the adiabatic lapse rate $\Gamma_d = g / c_p$

Today: What happens when vapour condenses? How do winds blow? Why has Earth's climate been stable for 4 Gyr?

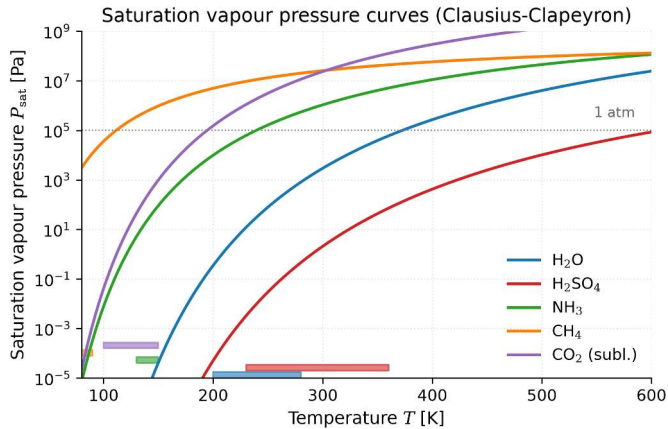


1. Cloud formation

Methane clouds over Titan, Cassini ISS. Credit: NASA/JPL-Caltech/SSI

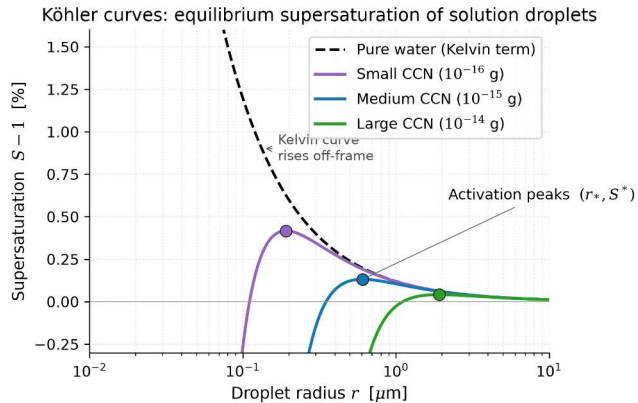
Saturation and condensation

- Every vapour has a **saturation vapour pressure** $P_{\text{sat}}(T)$ that rises roughly exponentially with temperature
- **Relative humidity** $\text{RH} = P_{\text{vapour}} / P_{\text{sat}}(T) \times 100\%$: saturated at 100%, **supersaturated** above it
- Rising air cools, P_{sat} drops, the air reaches saturation: clouds



Saturation vapour pressure against temperature for the five main condensables. Course figure.

One equation per species, combined with the T profile, governs every cloud deck in the solar system.



Köhler curves: the equilibrium supersaturation of solution droplets against droplet radius.
Course figure.

Condensation needs a **seed**: homogeneous nucleation from vapour alone needs $RH \gg 100\%$; heterogeneous nucleation on **condensation nuclei** (dust, sea salt, soot) activates at only 0.04–0.4% supersaturation.



A cumulonimbus anvil spreading at the tropopause. Credit: Eulenjäger, CC BY-SA 3.0.

Rising air cools at $\Gamma_d = g/c_p$ until it reaches its **dew point** T_d : the **lifting condensation level** (LCL) marks the **cloud base**, and above it latent heat lowers it to the **moist adiabatic lapse rate**, about 5 K km^{-1} (9.8 dry).

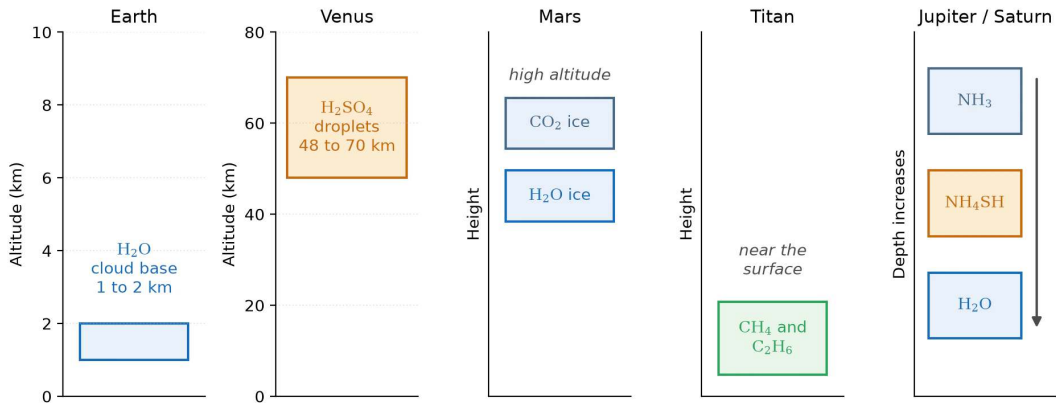
Cloud types of the solar system

What condenses depends on temperature and composition:

Body	Cloud species	Cloud location
Earth	H ₂ O	~1–2 km
Venus	H ₂ SO ₄	48–70 km
Mars	CO ₂ ice, H ₂ O ice	10–100 km
Titan	CH ₄ , C ₂ H ₆	~8–30 km
Jupiter	NH ₃ , NH ₄ SH, H ₂ O	layered, 0.5–7 bar

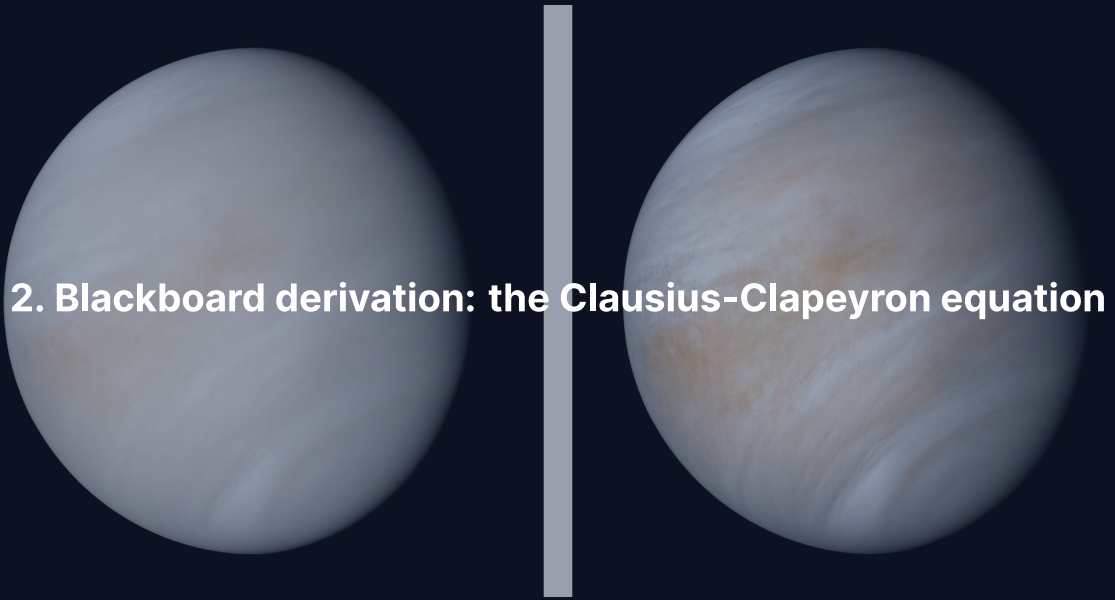
The **Clausius-Clapeyron equation** governs every cloud deck.

Which species condenses where



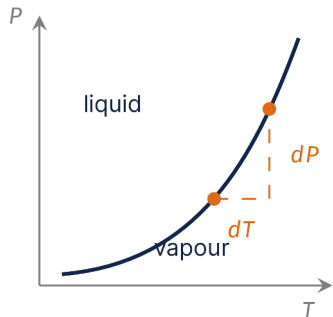
the same Clausius-Clapeyron physics in every case

Which species condenses where: H₂O on Earth, H₂SO₄ on Venus, CO₂ and H₂O ice on Mars, CH₄ and C₂H₆ on Titan, the NH₃, NH₄SH and H₂O decks on Jupiter and Saturn. Course schematic.



2. Blackboard derivation: the Clausius-Clapeyron equation

Goal and strategy



Goal: the exponential dependence of P_{sat} on temperature from phase equilibrium.

- Along the coexistence curve $g_\ell(T, P) = g_v(T, P)$, so a step (dT, dP) keeps $dg_\ell = dg_v$
- Read off the slope dP/dT and insert the latent heat
- Approximate and integrate

To the board.

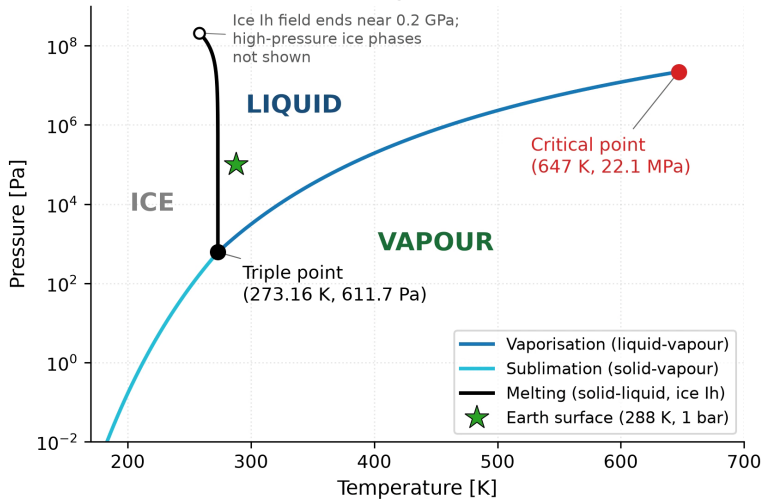
The Clausius-Clapeyron equation

$$P_{\text{sat}}(T) = P_{\text{ref}} \exp \left[-\frac{L_v}{R_v} \left(\frac{1}{T} - \frac{1}{T_{\text{ref}}} \right) \right]$$

Credit: Derivation follows Catling & Kasting (2017), *Atmospheric Evolution on Inhabited and Lifeless Worlds*, Ch. 1

- P_{sat} depends **exponentially** on temperature
- The sensitivity is set by L_v/R_v (units of K): large values mean a steep curve
- The equation applies to **any** vapour to condensed-phase transition

Phase diagram of water



Phase diagram of water: the liquid-vapour curve integrates Clausius-Clapeyron from the triple point to the critical point, the solid-vapour curve governs sublimation. Course figure.

Worked example: water on Earth

For water vapour:

L_v	$2.50 \times 10^6 \text{ J kg}^{-1}$
M	$0.018 \text{ kg mol}^{-1}$
$R_v = R^* / M$	$462 \text{ J kg}^{-1} \text{ K}^{-1}$
T_{ref} (triple point)	273 K
P_{ref} (triple point)	611 Pa
L_v / R_v	$\approx 5400 \text{ K}$

At $T = 293 \text{ K}$ (20°C):

$$P_{\text{sat}} = 611 \exp\left[-5400\left(\frac{1}{293} - \frac{1}{273}\right)\right] = 611 \times 3.86 \approx 2360 \text{ Pa}$$

Measured value: 2300 Pa; excellent agreement.

Condensing species in the solar system

Species	L_v (kJ kg ⁻¹)	R_v (J kg ⁻¹ K ⁻¹)	L_v/R_v (K)	T_{cond} (K)	Where
H ₂ O	2500	462	5400	200–280	Earth, Mars
H ₂ SO ₄	540	85	6400	230–360	Venus
NH ₃	1370	488	2800	130–150	Jupiter, Saturn
CH ₄	510	519	980	80–90	Titan, Uranus, Neptune
CO ₂ [†]	571	189	3020	100–150	Mars

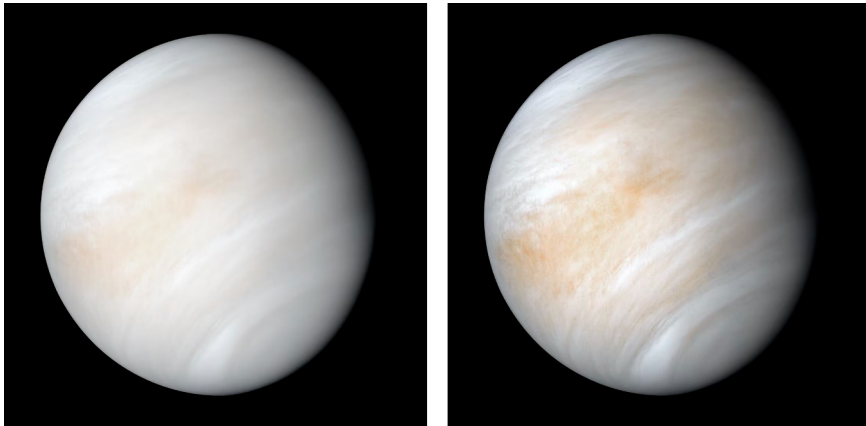
[†]latent heat of sublimation, solid to vapour

Cloud-base sharpness scales with $(L_v/R_v) \Gamma/T^2$: Venus's lapse rate of 8 K km⁻¹ sharpens its base; Titan's 1.3 K km⁻¹ spreads methane clouds over about 15 km.



3. Clouds of the solar system

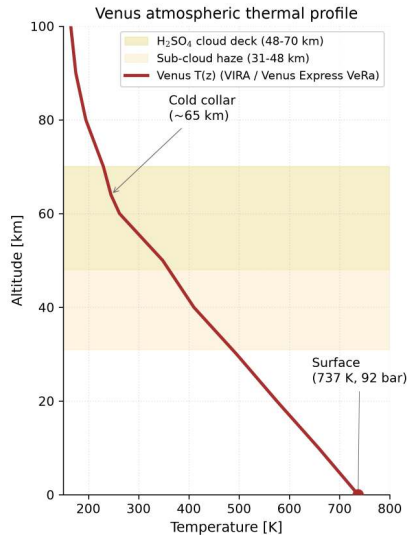
Water-ice clouds over Mars, Mars Odyssey THEMIS (2025). Credit: NASA/JPL-Caltech/ASU



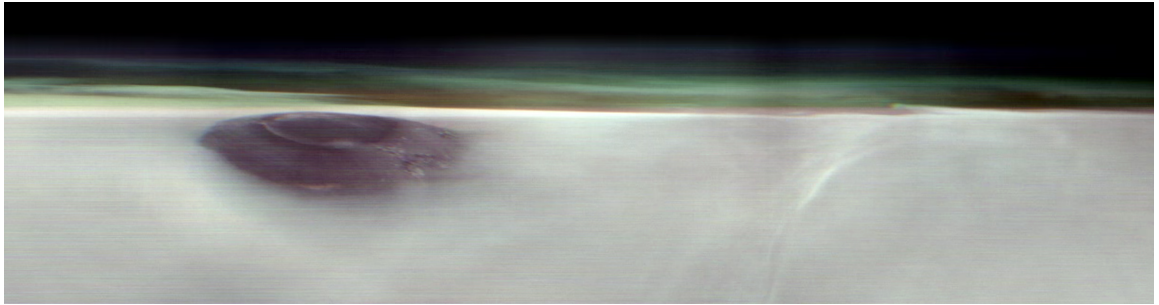
Venus in ultraviolet light, where an unidentified absorber traces the upper cloud motions.

Credit: NASA/JPL-Caltech (1974).

Concentrated H_2SO_4 clouds span 48–70 km, made by **photochemistry** from outgassed SO_2 ($\text{SO}_2 + \text{O} \rightarrow \text{SO}_3$, $\text{SO}_3 + \text{H}_2\text{O} \rightarrow \text{H}_2\text{SO}_4$); a sub-cloud haze reaches down to about 31 km.

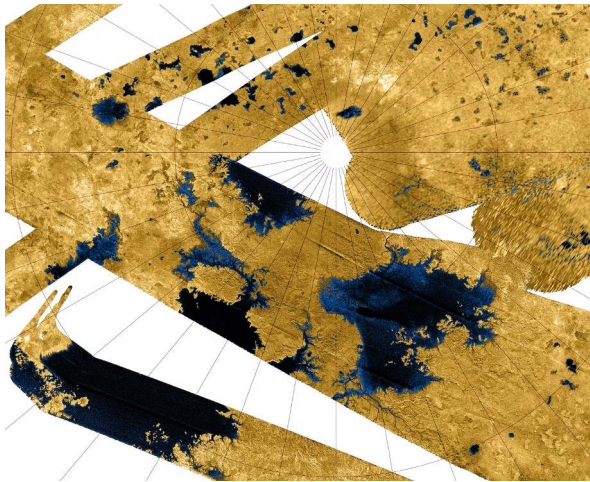


Venus thermal structure from VIRA and Venus Express VeRa: the 22 km H₂SO₄ deck sits at 48–70 km, where Clausius-Clapeyron allows it. Course figure.



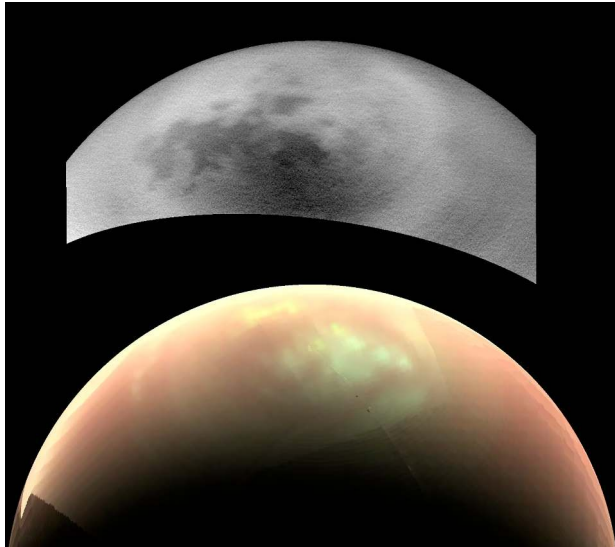
Water-ice clouds over the Tharsis limb, Mars Odyssey THEMIS, 2 May 2025. Credit: NASA/JPL-Caltech/ASU.

CO_2 ice clouds form at 50–100 km where T falls below the CO_2 frost point (148 K at the 6 mbar surface, lower aloft), H_2O ice clouds at 10–30 km over Tharsis, both seeded by mineral dust.

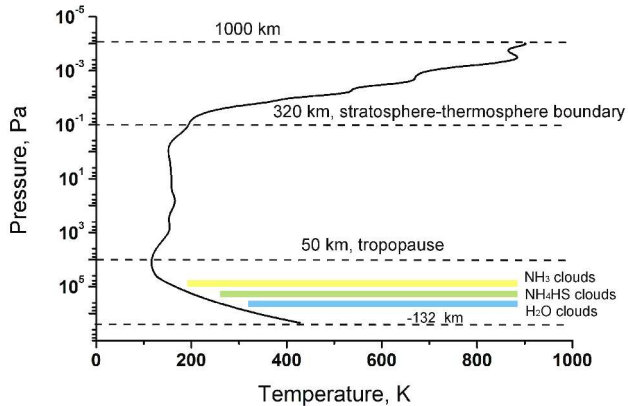


Titan's polar lakes and seas in Cassini radar. Credit: NASA/JPL-Caltech/ASI/USGS.

At 94 K and 1.5 bar methane sits near its triple point: rain carves channels and fills polar seas such as **Kraken Mare**.



Methane clouds at Titan's mid-southern latitudes, Cassini ISS: the only active hydrological cycle beyond Earth. Credit: NASA/JPL-Caltech/SSI.



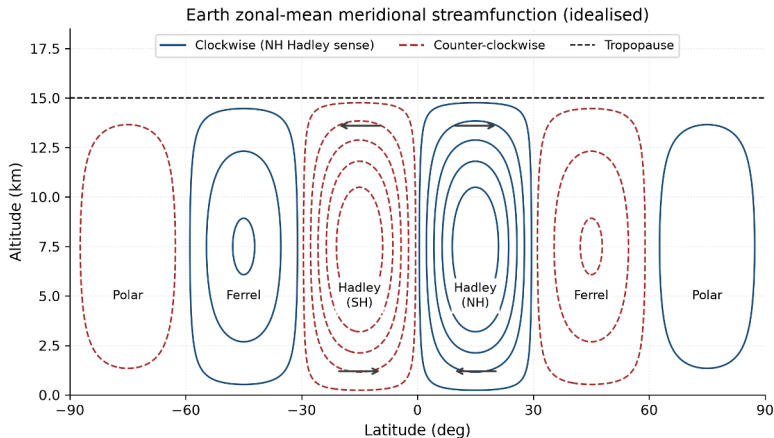
Jupiter's layered cloud structure. Credit: Wikimedia, CC BY-SA 3.0.

Clausius-Clapeyron predicts the stack: NH₃ ice at 130–150 K (0.5–1 bar), NH₄SH at 200–240 K (2–3 bar), H₂O at 270–300 K (5–7 bar), whose latent heat fuels moist convection.

A satellite image of Earth from space, showing a wide view of the planet's surface. The image captures the curvature of the Earth, with a clear blue sky and a dark, silty horizon line. The surface is covered with a complex pattern of white and grey clouds, which appear to be swirling and flowing in a way that suggests atmospheric dynamics. The clouds are more concentrated in some areas, particularly in the upper right, and more sparse in others. The overall color palette is dominated by the deep blue of the sky and the white and grey of the clouds, with some darker, more textured areas visible on the ground surface.

4. Atmospheric dynamics

Jet-stream cirrus from orbit. Credit: NASA/JSC



Idealised zonal-mean overturning streamfunction of Earth's troposphere, after Held & Hou (1980). Course figure.

Equatorial heating lifts air that flows poleward aloft, sinks near 30° latitude and returns as the **trade winds**.

The Coriolis effect

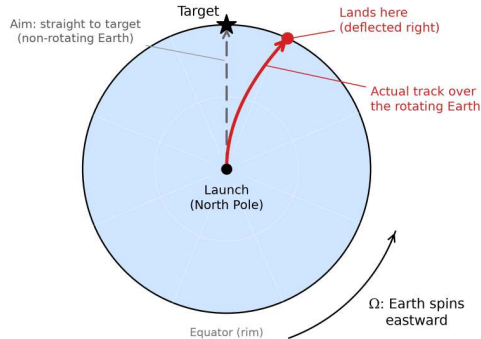
Moving air is deflected right in the Northern Hemisphere, left in the Southern.
The **Coriolis parameter**:

$$f = 2\Omega \sin \phi$$

- Earth: $\Omega = 7.29 \times 10^{-5} \text{ rad s}^{-1}$
- Mid-latitudes ($\phi = 45^\circ$): $f \approx 1.03 \times 10^{-4} \text{ s}^{-1}$
- Equator: $f = 0$, no deflection

Winds circulate around pressure centres instead of blowing from high to low.

Geometric origin of the Coriolis deflection
(view from above the North Pole)



The Earth turns east under the flight, so the track bends to the right of the direction of travel (Northern Hemisphere; it bends left in the Southern).

Coriolis deflection of a polar launch seen from above the North Pole: an apparent force from the planet rotating beneath a straight path. Course figure.

The Rossby number

Does planetary rotation dominate a given flow?

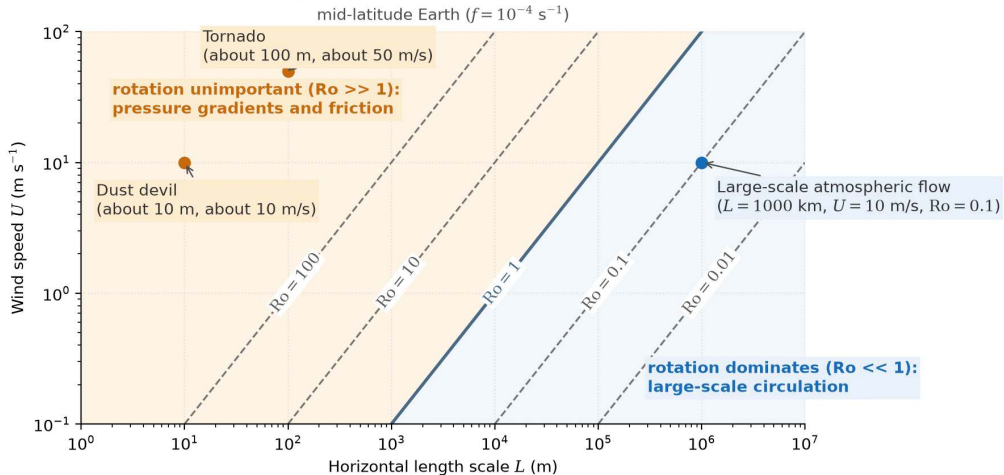
$$Ro = \frac{U}{fL}$$

$Ro \ll 1$: rotation dominates: large-scale circulation, jet streams (Earth:
 $U \sim 10 \text{ m s}^{-1}$, $L \sim 1000 \text{ km}$, $Ro \sim 0.1$)

$Ro \gg 1$: rotation unimportant: convective plumes, tornadoes

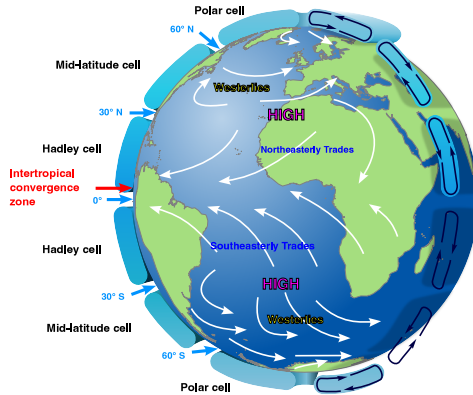
The Rossby number sets the dynamic regime and the number of circulation cells.

The Rossby number sorts flows by the role of rotation



Flow regimes sorted by $Ro = U/(fL)$ for mid-latitude Earth: large-scale circulation where rotation dominates, dust devils and tornadoes where pressure gradients and friction govern.

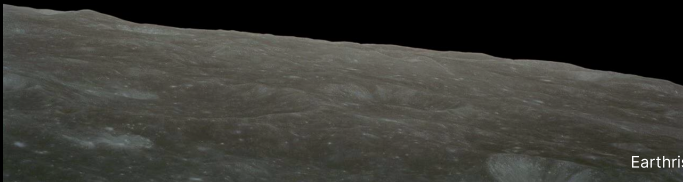
Course figure.



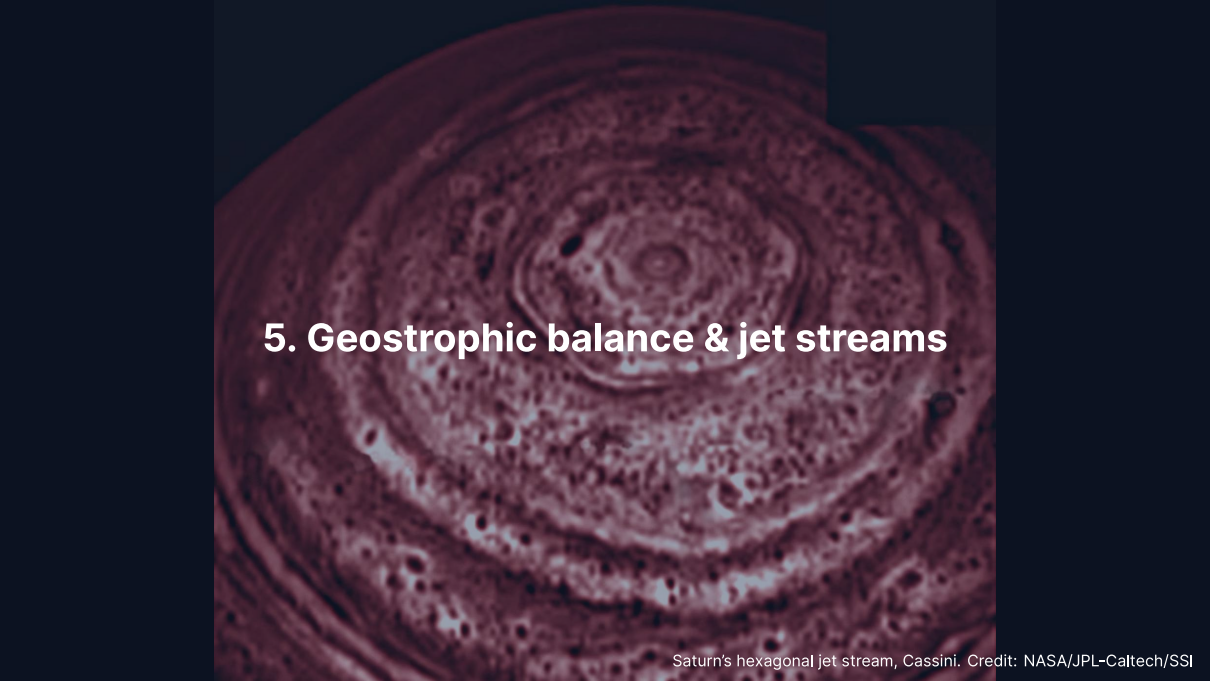
Earth's three-cell circulation, the pattern of a moderate rotator: Hadley, mid-latitude (Ferrel) and polar cells. Credit: Wikimedia Commons, public domain.

Slow rotators (Venus, Titan): one equator-to-pole cell per hemisphere; Earth: three cells (Hadley, Ferrel, polar); **fast rotators** (Jupiter, Saturn): many cells that appear as banded jets.

Break



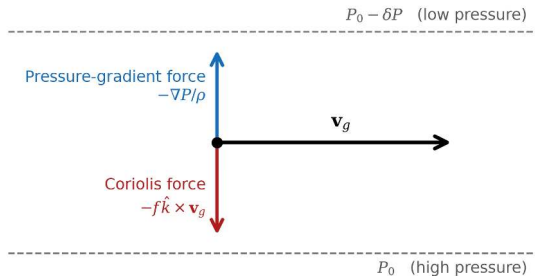
Earthrise, Apollo 8 (1968). Credit: NASA

A high-resolution image of Saturn's atmosphere showing a prominent hexagonal jet stream. The hexagon is a large, six-sided cloud pattern with a central vortex. The surrounding atmosphere shows various other cloud features and bands. The image is in a reddish-brown color palette.

5. Geostrophic balance & jet streams

Saturn's hexagonal jet stream, Cassini. Credit: NASA/JPL-Caltech/SSI

Geostrophic balance (Northern Hemisphere)



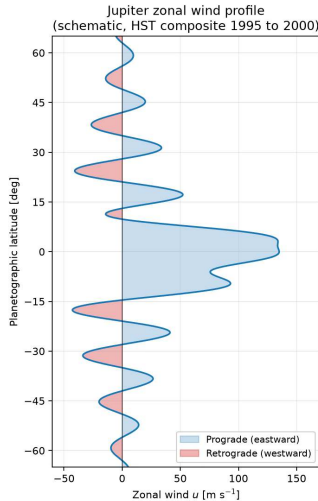
Geostrophic balance between the pressure-gradient force and the Coriolis force. Course figure.

At $R_o \ll 1$, $f\hat{k} \times \mathbf{v}_g = -\nabla P/\rho$: geostrophic winds blow **parallel to isobars**, cyclonically around lows and anticyclonically around highs.

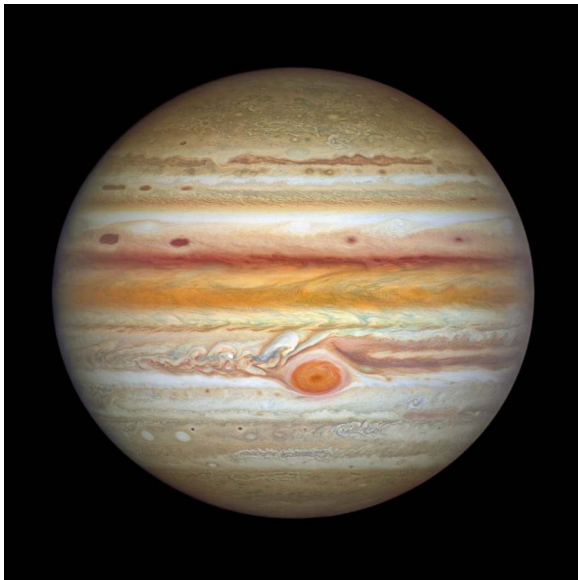


Jet-stream cirrus from orbit. Credit: NASA JSC, public domain.

The thermal wind, $\partial \mathbf{v}_g / \partial z \propto \hat{k} \times \nabla T$: the equator-to-pole temperature gradient drives $30\text{--}70 \text{ m s}^{-1}$ jets aloft, subtropical near 30° and polar near 60° , that steer mid-latitude weather.



Jupiter's cloud-top zonal winds: alternating prograde and retrograde jets peak near 180 m s^{-1} .
Course figure, after García Melendo & Sánchez-Lavega (2001).

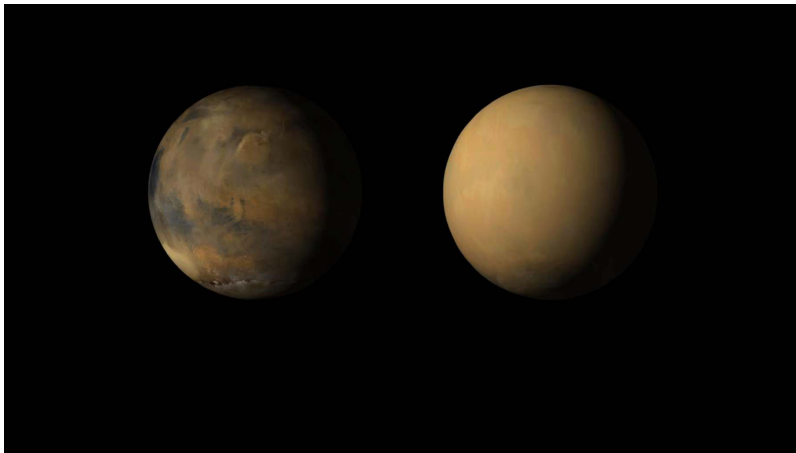


Jupiter's global map from the Hubble OPAL programme. Credit: NASA/ESA/STScI.

Bright **zones** are anticyclonic upwelling with cool ammonia ice tops, dark **belts** cyclonic downwelling exposing warmer layers; zonal jets flank the boundaries (Jupiter ~ 180 , Saturn $\sim 400 \text{ m s}^{-1}$); Juno gravity shows the jets reach about 3000 km deep (Kaspi et al. 2018).

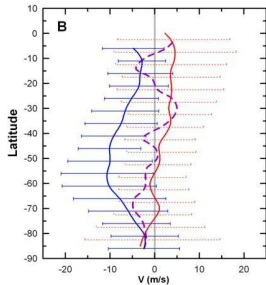
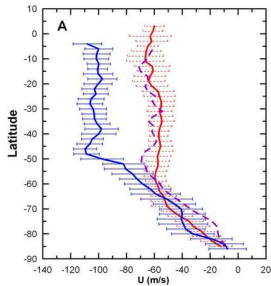
A close-up, high-resolution image of Jupiter's Great Red Spot, a massive, oval-shaped storm system. The storm is characterized by a reddish-orange hue and complex, swirling cloud patterns in shades of white, grey, and brown. The image is presented as a semi-circular arc against a dark blue background.

6. Weather & storms in the solar system



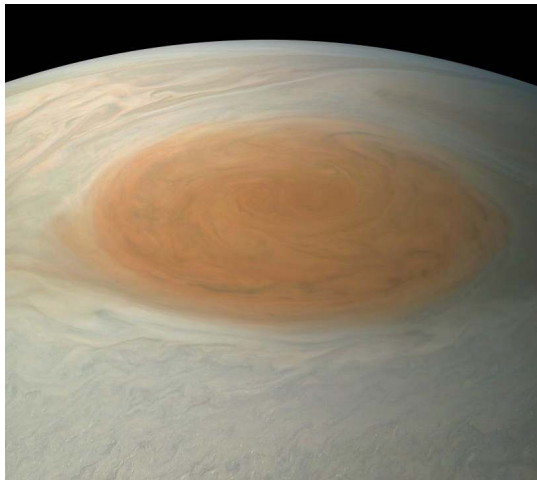
The 2018 global dust storm on Mars. Credit: NASA/JPL-Caltech/MSSS.

Dust lofted to 10–30 km absorbs sunlight, heats the air and strengthens the winds that loft more dust, a **positive feedback**; seasonal CO₂ frost on the caps (25–30% of the atmosphere) swings the surface pressure between about 5 and 7 mbar.



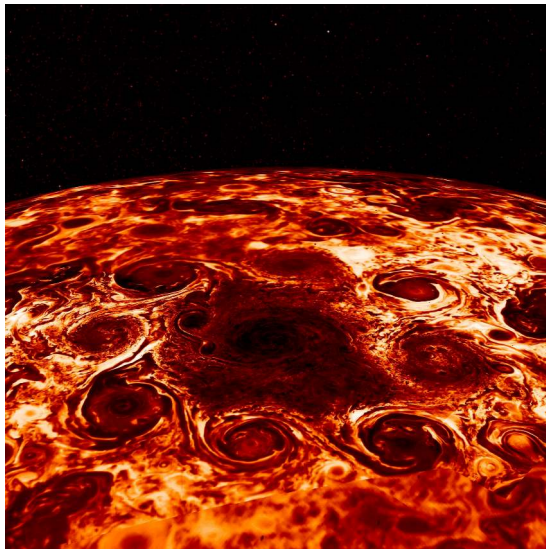
Venus cloud-level winds from VIRTIS on Venus Express: zonal (A) and meridional (B) speed against latitude. Sánchez-Lavega et al. (2008).

Cloud-top winds of $\sim 100 \text{ m s}^{-1}$ circle Venus in about 4 days, 60 times faster than the surface: **thermal tides** pump momentum upward and **planetary waves** carry it equatorward (Lecture 9).

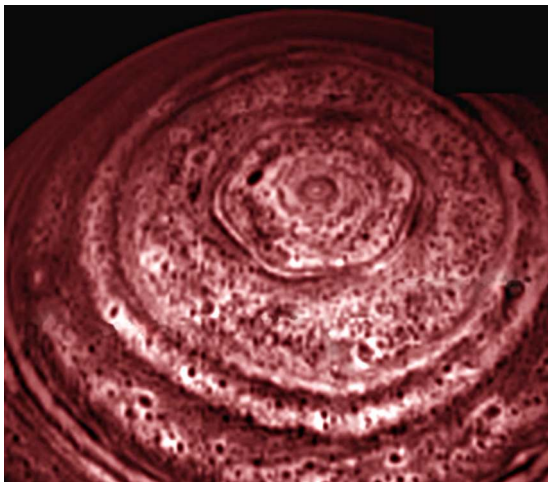


The Great Red Spot from Juno. Credit: NASA/JPL-Caltech/SwRI/MSSS.

An anticyclone larger than Earth, seen since about 1830, with winds near 120 m s^{-1} , fed by merging vortices and latent heat.



Jupiter's north polar cyclone cluster from Juno JIRAM: a central cyclone ringed by eight circumpolar cyclones in a stable octagon. Credit: NASA/JPL-Caltech/SwRI/ASI/INAF/JIRAM.



Saturn's north polar hexagon, Cassini 2013. Credit: NASA/JPL-Caltech/SSI.

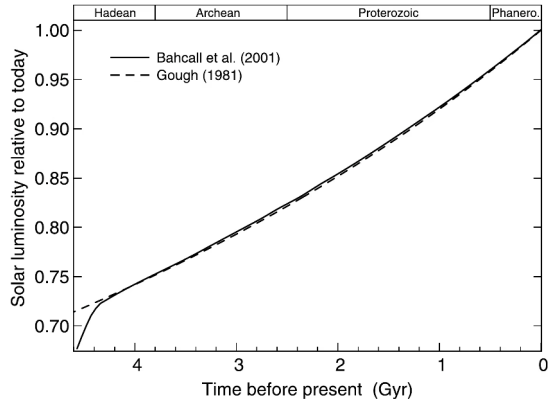
A stationary Rossby wave ($m = 6$) locked to the jet at $\sim 78^\circ\text{N}$, about 30,000 km across, seen by Voyager and Cassini.



Neptune's Great Dark Spot, Voyager 2 (1989): Neptune radiates 2.6 times the sunlight it absorbs, and that internal heat drives $\sim 400 \text{ m s}^{-1}$ winds. Credit: NASA/JPL-Caltech.

The background of the slide is a full-page image of a 'Snowball Earth' artist's impression. It shows a pale blue, icy planet with some darker, brownish-grey landmasses visible through the ice. The planet is set against a dark, starry space background. The title text is overlaid on the center of the planet.

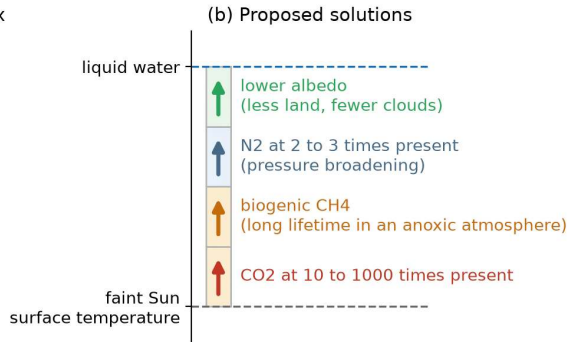
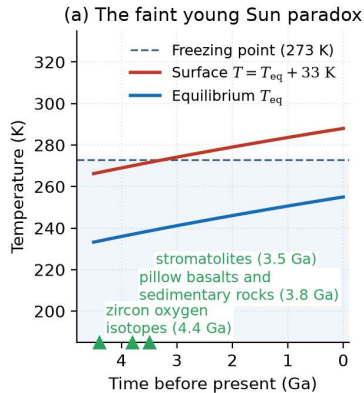
7. Climate evolution & the faint young Sun



Solar luminosity through geologic time: the Sun was about 30% fainter ($0.71 L_{\odot}$) at $t = 0$.

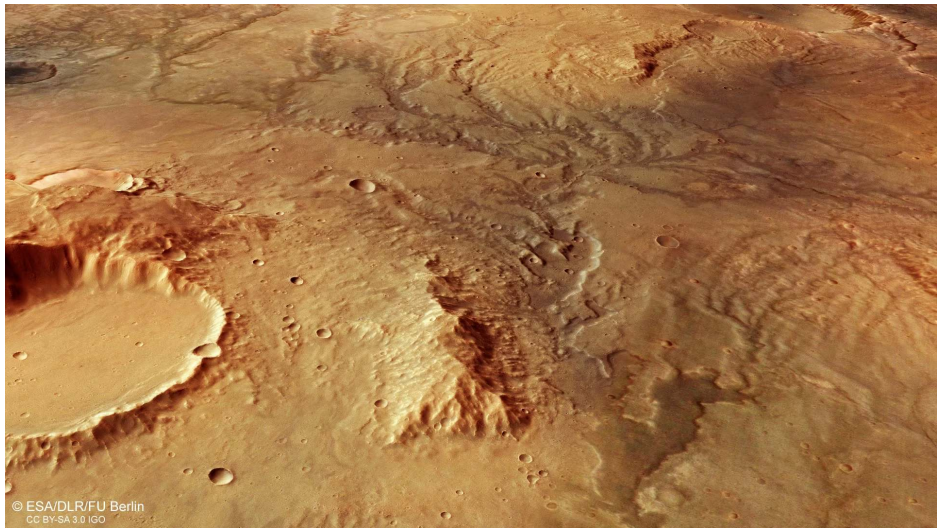
Reproduced from Feulner (2012).

$L(t)/L_{\odot} \approx [1 + \frac{2}{5}(1 - t/t_{\odot})]^{-1}$ gives $T_{\text{eq}} \approx 234$ K for the early Earth, yet zircons (4.4 Ga), pillow basalts (3.8 Ga) and stromatolites (3.5 Ga) record liquid water: the **faint young Sun paradox**.

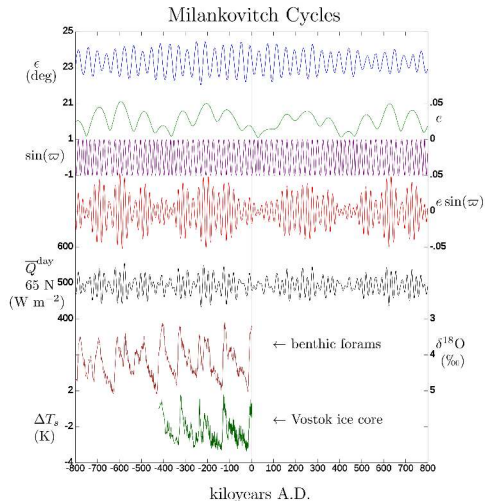


(a) Equilibrium temperature and surface temperature (33 K greenhouse warming added) under the fainter Sun against the geological record of liquid water; (b) the proposed warming contributions. Course figure.

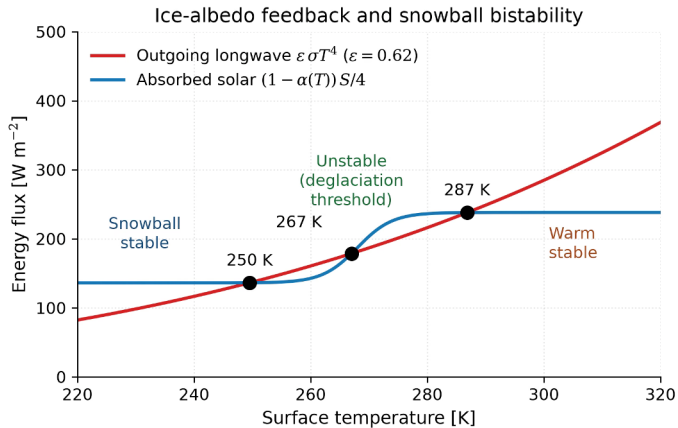
Solutions: CO₂ at 10 to 1000 times present, biogenic CH₄, more N₂, and a lower planetary albedo.



Noachian valley networks, Mars Express HRSC: warm, wet early Mars (>3.7 Ga) is an open paradox, since pure CO₂ condenses into bright clouds. Credit: ESA/DLR/FU Berlin, CC BY-SA 3.0 IGO.



Milankovitch cycles over 800 kyr: obliquity, eccentricity and precession pace the glacial cycles, though the 100 kyr rhythm needs non-linear feedbacks. Credit: Wikimedia Commons, CC BY-SA.



Energy-balance bistability of the ice-albedo feedback: stable states at 250 K and 287 K, an unstable threshold at 267 K. Course figure.

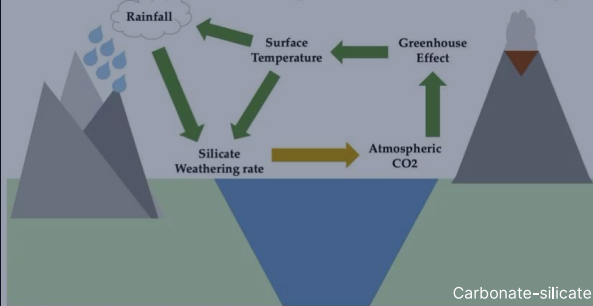
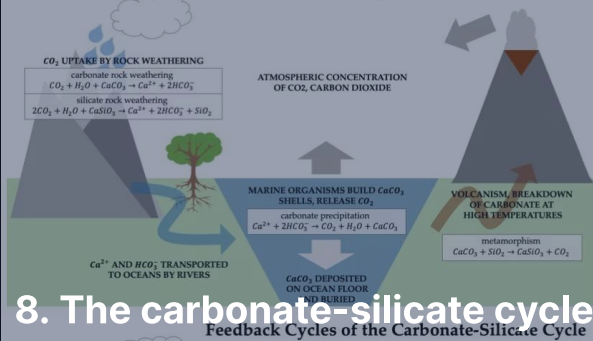
The **ice-albedo feedback** is positive: cooling expands reflective ice, and past $\sim 30^\circ$ latitude it runs away.



Snowball Earth, artist's impression. Credit: Oleg Kuznetsov, CC BY-SA 4.0.

Two Neoproterozoic glaciations (~717 and 635 Ma) left tropical glacial deposits at surfaces below -50°C ; ice shut down silicate weathering until volcanic CO_2 built up over 10–100 Myr and melted it (Hoffman et al. 2017).

The Geologic Carbonate-Silicate Cycle



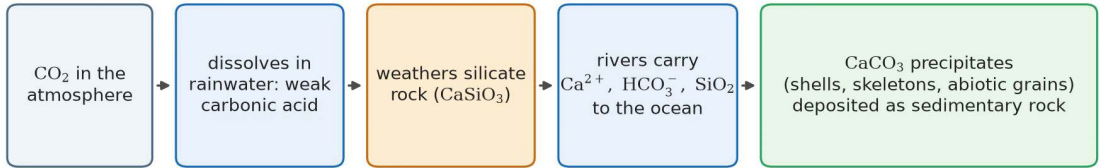
The Urey reaction

Silicate weathering, with water as the medium, draws atmospheric CO₂ into carbonates:



- CO₂ dissolves in rain as carbonic acid and weathers silicates
- Rivers carry Ca²⁺ and HCO₃⁻ to the ocean, where CaCO₃ precipitates
- **Plate tectonics** closes the loop: subduction and volcanism return the CO₂

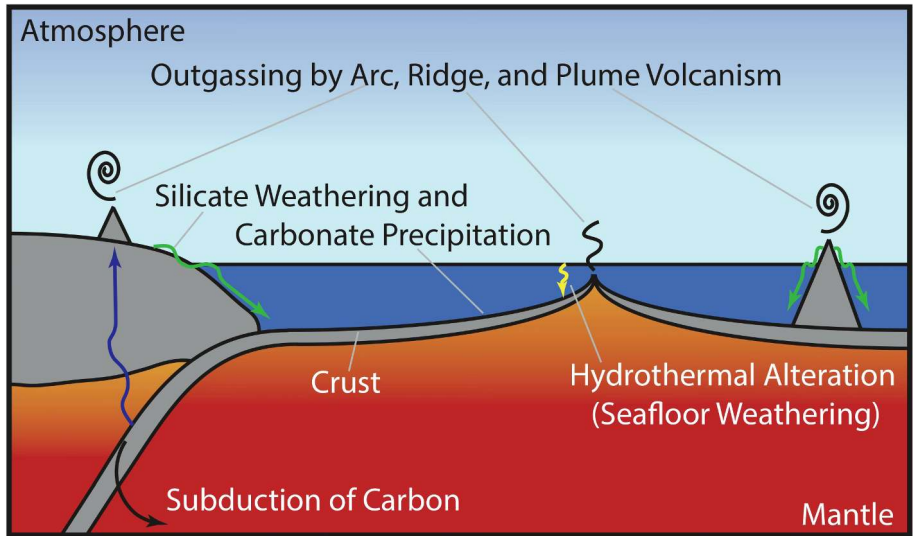
The Urey reaction pathway



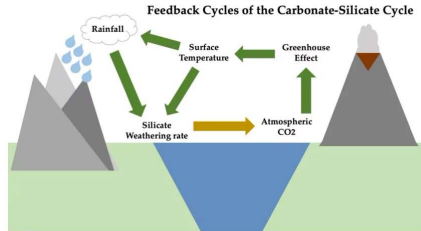
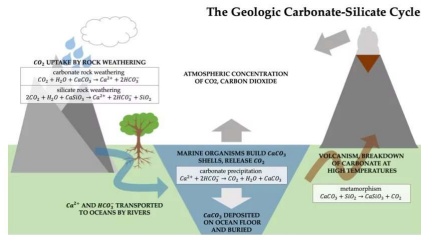
net effect: CO₂ drawn out of the atmosphere into carbonate rock, the long-term carbon sink



The Urey reaction pathway from rain to carbonate rock: CO₂ drawn out of the atmosphere into the long-term carbon sink. Course schematic.



The carbonate-silicate cycle on the modern Earth: weathering into oceanic carbonates, subduction, and metamorphic CO_2 returned by arc and plume volcanism. Credit: Foley (2024).



The carbonate-silicate cycle. Credit: Wikimedia, CC BY-SA 4.0.

Volcanic outgassing returns subducted CO₂: the steady state balances the weathering sink against the volcanic source.

The negative feedback thermostat

Silicate weathering rates rise exponentially with temperature:

$$W(T) = W_0 \exp\left(\frac{T - T_0}{T_e}\right)$$

- **Warming:** more rain and faster reactions draw CO_2 down and cool the planet
- **Cooling:** weathering slows and volcanic CO_2 accumulates
- With $T_e \approx 10$ K, doubling the weathering rate needs ~ 7 K of warming

Perturbations damp on ~ 0.5 Myr timescales, maintaining liquid water for >4 Gyr.

Source: Walker et al. (1981); Pierrehumbert (2010)

The thermostat needs both ingredients

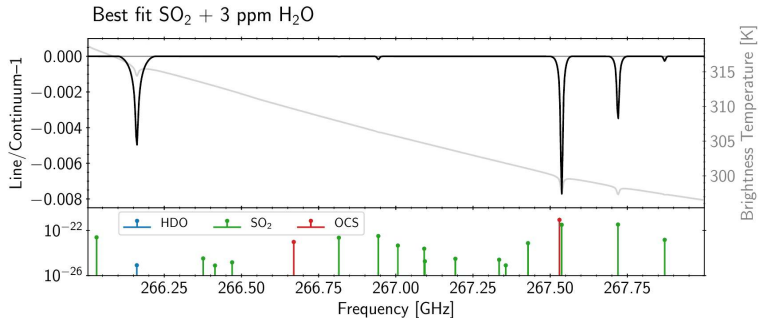
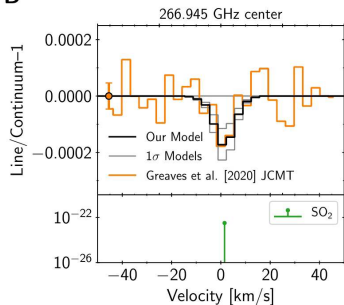
	Liquid water: yes	Liquid water: no
Active volcanism: yes	Earth Thermostat works CO ₂ regulated over Gyr	Venus Weathering stops, volcanic CO ₂ accumulates: 92 bar
Active volcanism: no	Mars (volcanism fading, 3 Ga) Source broken, weathering and escape draw CO ₂ down: 6 mbar	No cycle (Mars today) No water or volcanism Thermostat fails

The thermostat needs liquid water and volcanism: Earth has both, Venus lost its water, Mars its volcanism and then its water. Course schematic.

Venus lost the sink (92 bar CO₂), Mars lost the source and then its water: habitability needs active geological cycling.

The image features two spheres against a dark blue background. The sphere on the left is detailed with various shades of brown, tan, and blue, representing the surface features of Mars. The sphere on the right is a uniform, hazy light brown color, representing Mars during a global dust storm. The text '9. Recent advances & outlook' is centered between the two spheres.

9. Recent advances & outlook

D

The reported 267 GHz Venus phosphine feature (left: line contrast against Doppler velocity, JCMT data in orange) is fit by mesospheric SO₂ (right: line contrast and brightness temperature against frequency, molecular line positions below), a controversy that NASA's DAVINCI probe will test in situ. Reproduced from Lincowski et al. (2021), Fig. 3D.

Summary

1. The Clausius-Clapeyron equation predicts saturation vapour pressures and cloud bases on every planet
2. Rotation sets the circulation: Coriolis, Rossby number, geostrophic winds and jets, from single Hadley loops to banded jets
3. The faint young Sun paradox is resolved by greenhouse warming and lower albedo; carbonate-silicate weathering is a ~ 0.5 Myr climate thermostat

Next: Lecture 7

Questions?

Next: Lecture 7, Planetary surfaces: geology, geomorphology & geophysics

Mon 28 Sep, 15:00–17:00, room 5419.0161 (Collegezaal)

Next tutorial: Fri 25 Sep, 13:00–15:00, room 5419.0161 (Collegezaal)

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